



Anglo-Chinese Junior College

JC2 Biology Preliminary Examination

Higher 2



A Methodist Institution
(Founded 1886)

CANDIDATE
NAME

FORM
CLASS

TUTORIAL
CLASS

INDEX
NUMBER

BIOLOGY

Paper 2 Structured Questions

9744/02

25 August 2022

2 hours

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Name, Class and Index number in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiners' use only	
1	/ 9
2	/ 10
3	/ 11
4	/ 9
5	/ 10
6	/ 10
7	/ 10
8	/ 10
9	/ 10
10	/ 6
11	/ 5
Total	/ 100

Answer **all** questions.

- 1 Fig. 1.1 shows *Mycobacterium tuberculosis* infecting an alveolar cell in the lungs.

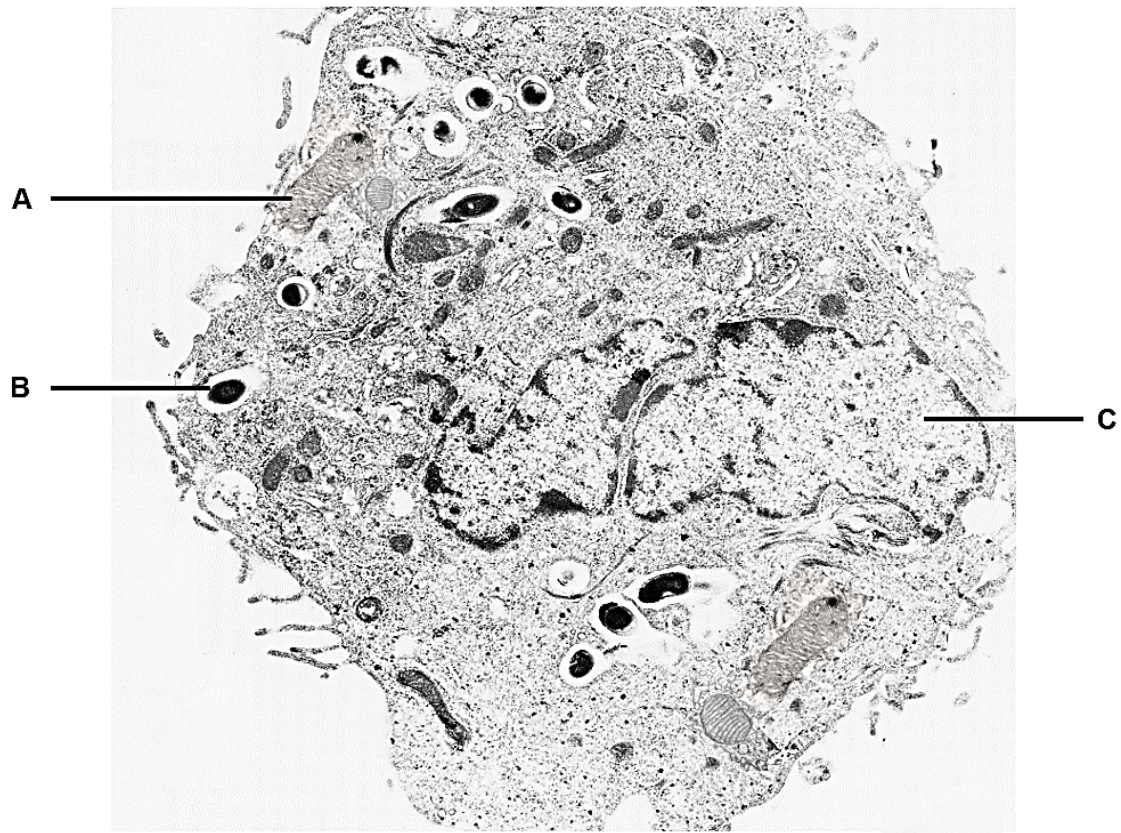


Fig. 1.1

- (a) (i) Using the letters **A** to **C**, identify the mitochondrion.

..... [1]

- (ii) Justify your answer to (a)(i).

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 [2]

- (b) Compare the structures of *M. tuberculosis* and a mitochondrion.

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- (c) Explain how the structure of organelle C allows it to serve its functions.

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[Total: 9]

[Turn over

- 2 The rate of carbon dioxide uptake at a range of carbon dioxide concentrations by barley and sugar cane were compared at two temperatures using the apparatus shown in Fig. 2.1.

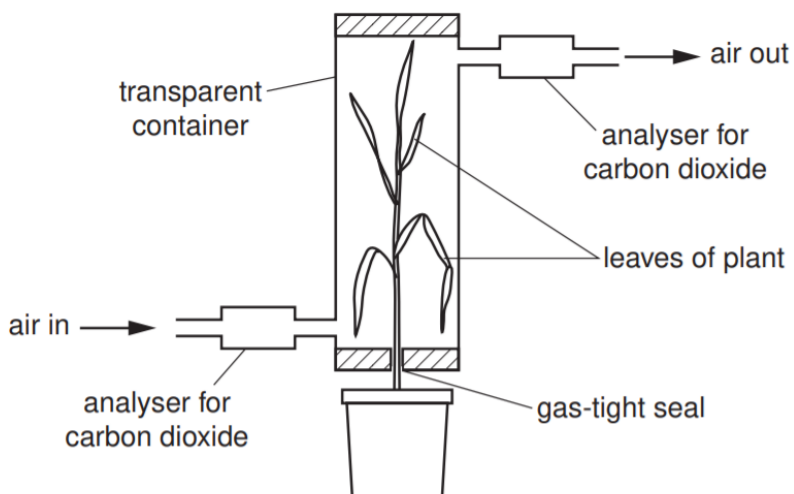


Fig. 2.1

The results of the experiment are presented in Fig. 2.2.

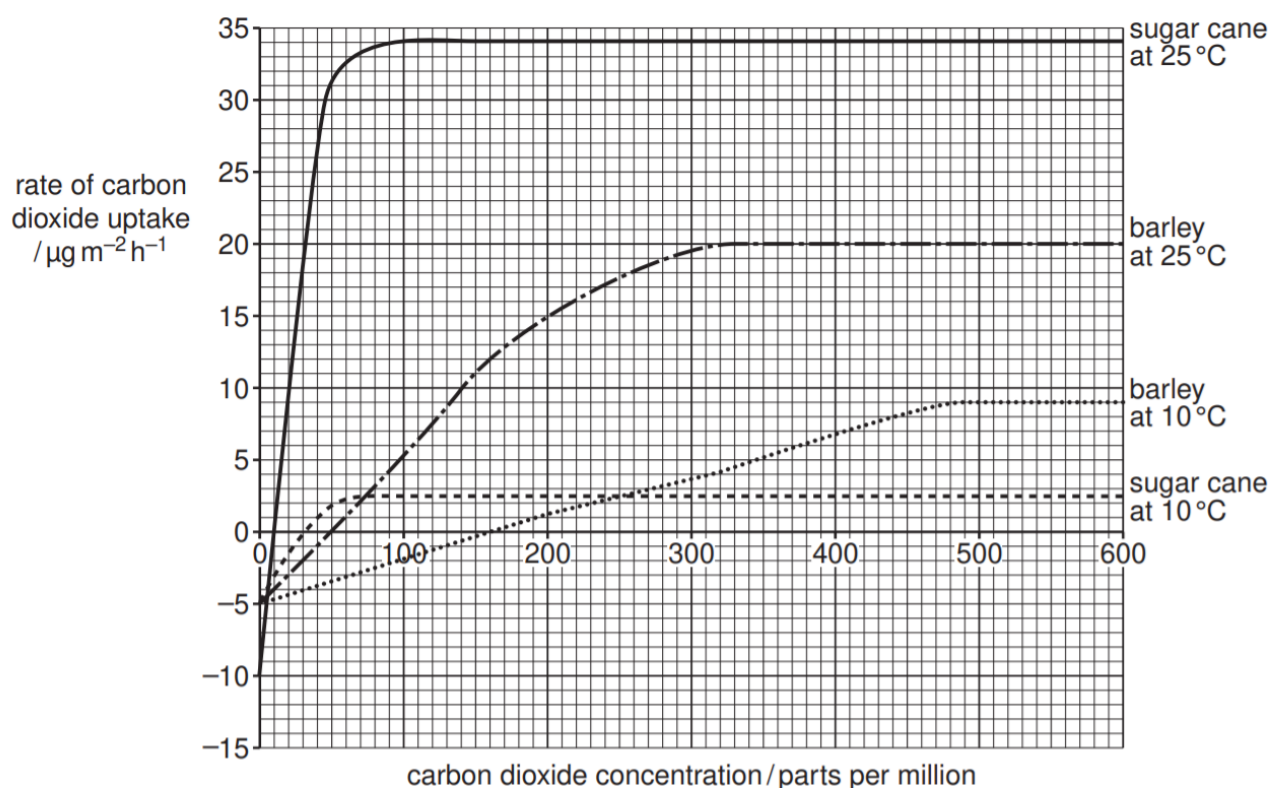


Fig. 2.2

- (a) (i) With reference to Fig. 2.2, describe the differences in rates of carbon dioxide uptake by barley and sugar cane in response to increasing carbon dioxide concentrations at 25°C.

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- (ii) Barley and sugar cane thrive in different climatic conditions. Complete Table 2.1 to show where these plants are likely to be found.

Table 2.1

climatic conditions	plant
tropical, warm climate	
temperate, cool climate	

[1]

[Turn over

Sugar cane is a cash crop which produces a high concentration of sucrose. Invertase found in the sugar cane hydrolyses sucrose into glucose and fructose, resulting in post-harvest loss of sucrose.

- (b)** Explain why a non-competitive inhibitor of invertase can potentially be used to reduce such losses.

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..... [2]

Starch is also found in sugar cane as a storage polysaccharide.

- (c) (i)** Describe the advantages to a plant of condensing glucose molecules into starch.

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- (ii)** Suggest why mammals store glucose as glycogen rather than as starch.

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[Total: 10]

- 3 Table 3.1 shows the telomerase activity and change in telomere length over time in different human stem cells.

Table 3.1

type of stem cell	telomerase activity	telomere length over time
germline	high	maintained
zygotic	high	maintained
embryonic	high	maintained
blood	low	shortened
skin	low	shortened
intestinal	low	shortened

- (a) (i) With the exception of germline stem cells, explain how the potency of stem cells is related to the telomere length in these cells.

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- (ii) Germline stem cells are the diploid cells that can only give rise to gametes, hence are considered unipotent.

Suggest why telomerase activity is high in germline stem cells.

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..... [1]

[Turn over

- (iii) Like telomeres, the centromere is an example of a non-coding DNA.

Compare the structures of telomeres and centromeres.

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The Hayflick limit is the number of times a cell population can divide before cell division stops, assuming an absence of telomerase activity.

Healthy human cells start off with an average telomere length of 10 kilobases (kb). Telomeres shorten at rates of 140 to 210 bases per replication prior to each cell division. It is estimated that when telomere lengths reach the threshold of 1.6 kb, the cells become senescent.

- (b) (i) Using the information provided, calculate the range of values for the Hayflick limit.

..... to divisions [2]

- (ii) Describe two possible deleterious effects on DNA when telomeres are depleted.

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..... [2]

[Total: 11]

- 4 One of the three tenets of the cell theory states that all living organisms are composed of one or more cells. Viruses challenge this tenet as they possess both living and non-living characteristics.

Giant viruses such as mimiviruses have been discovered recently. It is debatable whether such viruses are considered as living organisms or non-living entities. Mimivirus contains a DNA genome which is significantly larger than the genomes of any other known virus and is comparable to that of a cell. Like other viruses, it is dependent on its host for translation.

Fig. 4.1 shows the structure of Mimivirus.

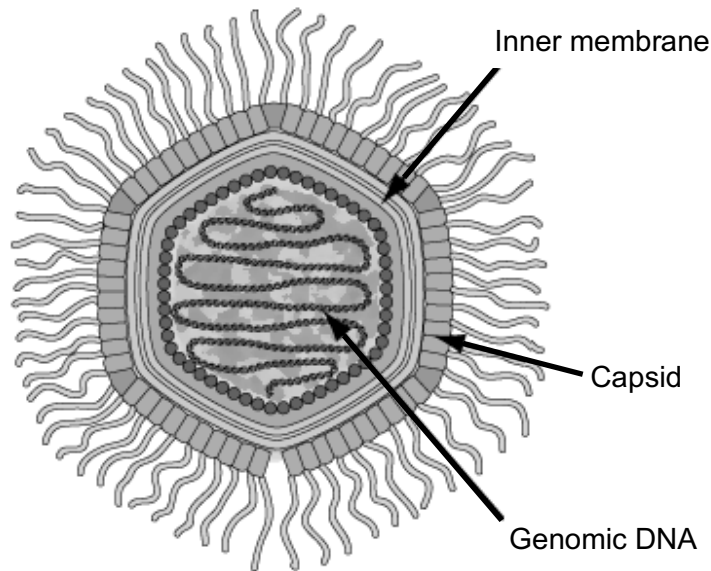


Fig. 4.1

- (a) With reference to the information provided, discuss if the discovery of mimivirus justify the classification of viruses as living.

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[Turn over

Fig. 4.2 shows the structure of an influenza virus.

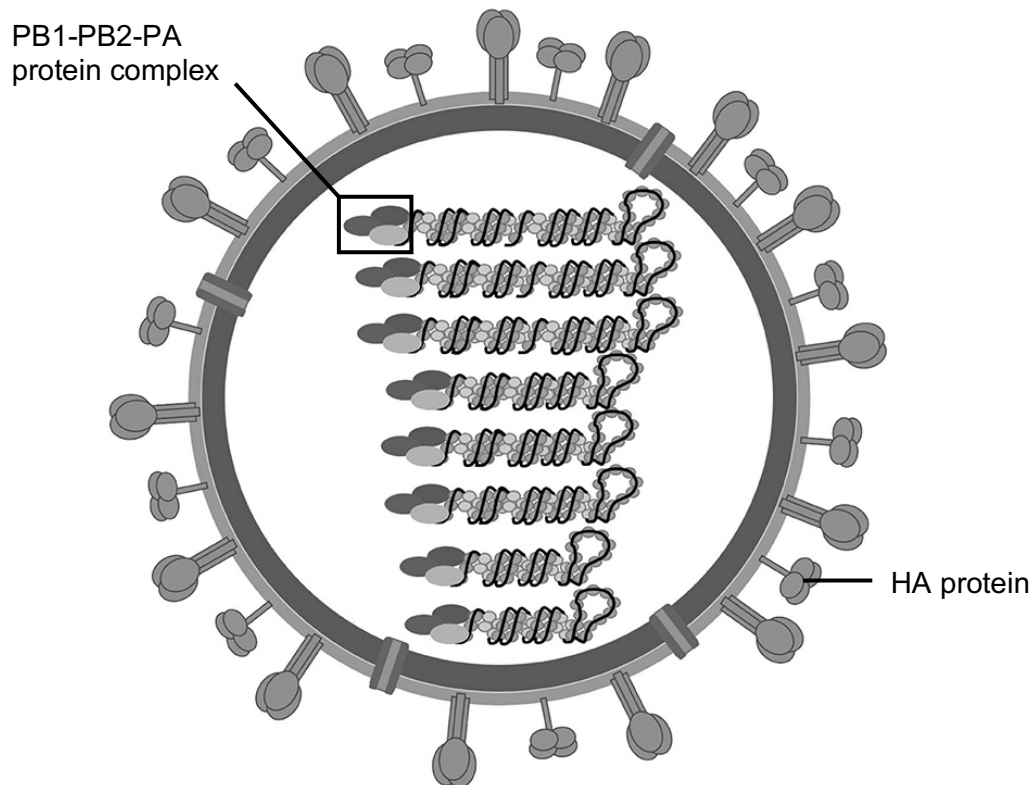


Fig. 4.2

(b) Describe the role of the following components labelled in Fig. 4.2 in the viral reproductive cycle:

(i) PB1-PB2-PA protein complex.

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..... [2]

(ii) HA protein.

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Oseltamivir is an orally administered antiviral medication that is recommended for patients infected with influenza virus. Fig. 4.3a shows the mechanism of action of oseltamivir and Fig. 4.3b shows its molecular structure as compared to that of sialic acid.

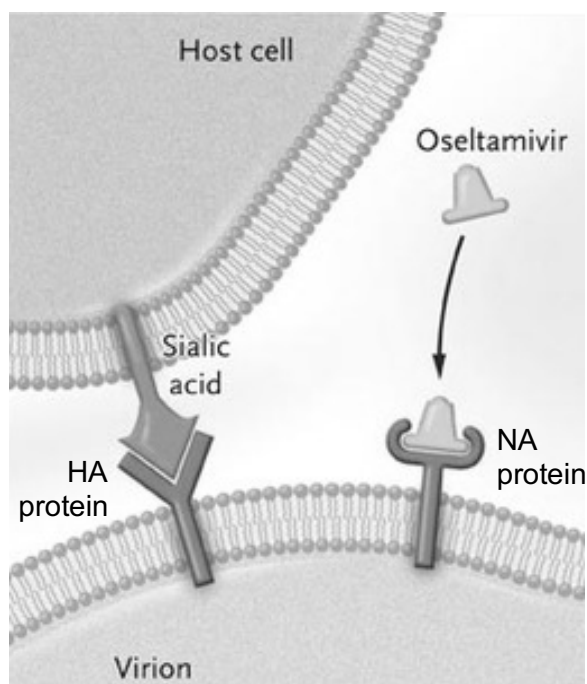


Fig. 4.3a

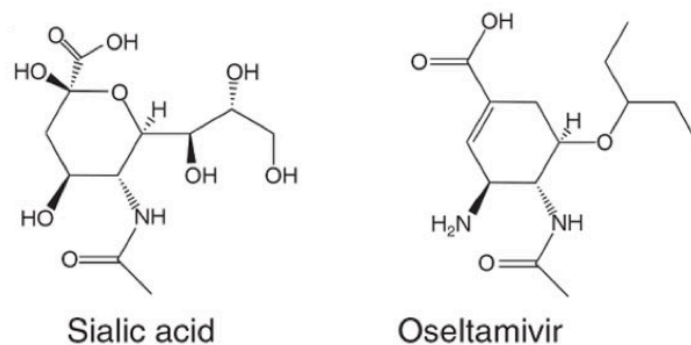


Fig. 4.3b

(c) Explain how oseltamivir affects the reproductive cycle of H1N1 virus.

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[Total: 9]

[Turn over

5 Fig. 5.1 shows the expression of the *lac* operon.

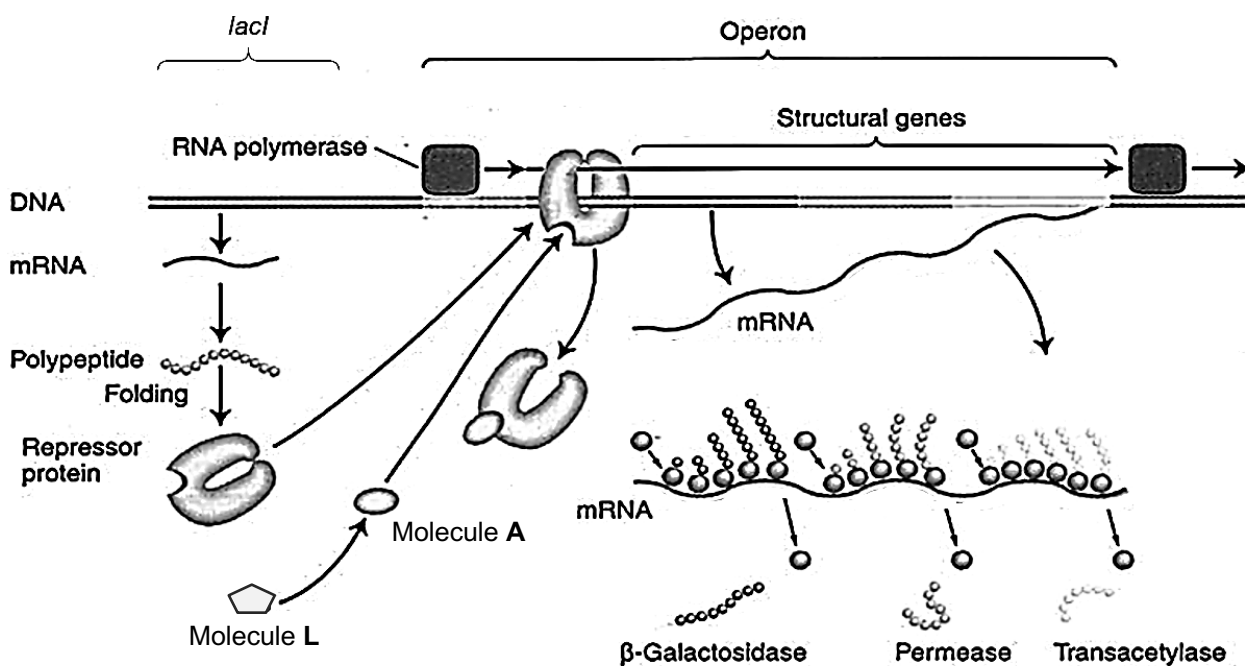


Fig. 5.1

(a) With reference to Fig. 5.1, identify molecule A.

..... [1]

An investigation was conducted to find out more about the *lac* operon. Bacteria were grown in a medium containing 2% glucose and 2% lactose. Three different experiments were conducted.

- experiment A – Wild type bacteria.
- experiment B – Bacteria with a mutation.
- experiment C – Wild type bacteria with IPTG added. IPTG is a molecular mimic of allolactose.

Fig. 5.2 shows the results of the investigation.

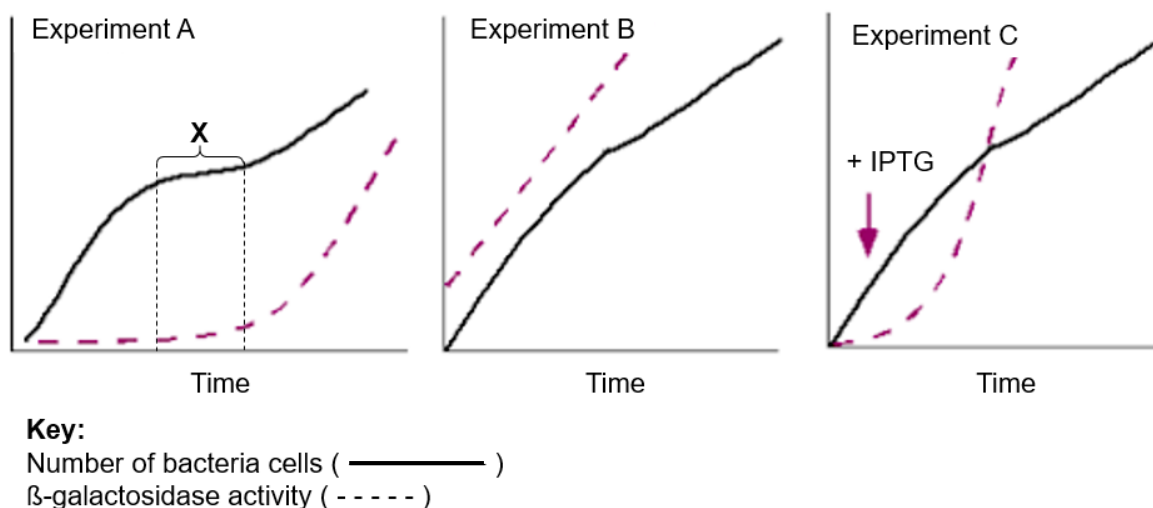


Fig. 5.2

(b) (i) Explain the bacterial growth during period **X** in experiment **A**.

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(ii) Suggest what mutation might have occurred in the bacteria used in experiment **B**.

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(iii) Explain the effect of IPTG in experiment **C**.

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[Turn over]

- (c) Explain how bacteriophages could enable more bacteria in a population to metabolise lactose.

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[Total: 10]

- 6 Cancer is a disease in which genetic alterations, most frequently chromosomal aberrations, are acquired in somatic cells. Fig. 6.1 shows a karyotype from a patient diagnosed with leukemia.



Fig. 6.1

- (a) (i) Describe two types of chromosomal aberrations evident in Fig. 6.1.

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..... [2]

[Turn over

- (ii) Acquired chromosome aberrations affecting specific genes are found to be associated with specific types of cancer, such as leukemia.

Explain how the chromosomal aberrations identified in **(a)(i)** could have resulted in uncontrolled division of cells in a leukemia patient.

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The effect of papaya peel extract on the ability of human tumour cells to divide was investigated. Different concentrations of papaya peel extract were added to cultures of human liver tumour cells, which were then incubated at 37°C. Table 6.1 shows the percentage of tumour cells that are still able to divide after incubation for 48 hours.

Table 6.1

papaya peel extract concentration/ $\mu\text{g cm}^{-3}$	percentage of tumour cells that are able to divide
0	100
5	63
10	58
20	46
40	40
80	37

- (b) (i) Suggest how the papaya peel extract could prevent the tumour cells from dividing by targeting specific phases of mitosis.

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..... [2]

- (ii) A student stated that $80 \mu\text{g cm}^{-3}$ of papaya peel extract would be suitable as a treatment for skin tumours in humans.

State the reasons why this statement is **not** supported by this study.

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..... [2]

[Total: 10]

[Turn over

- 7 Watermelon is a popular fruit crop. The colour intensity of watermelon flesh is a characteristic that some consumers look out for, as there is a perceived positive correlation of flesh colour intensity with sweetness.

Flesh colour intensity is controlled by gene **R/r**, in which allele **R** gives rise to scarlet red flesh (darker shade), while allele **r** gives rise to coral red flesh (lighter shade). Analysing the sequence of the alleles showed that allele **R** had a restriction site for the restriction enzyme *Hpa*II. This restriction site is absent in allele **r** due to a single nucleotide substitution.

Gene **T/t** codes for a sugar transporter which is responsible for different sweetness levels in the different cultivars.

The scientists then carried out an analysis of three generations of watermelon plants (I to III). The fruits produced by each plant were assessed for two features:

- the presence or absence of the *Hpa*II restriction site
- sweetness (the fruits were classified “sweet” or “not as sweet”)

Fig. 7.1 shows the pedigree of the three different generations of watermelon plants.

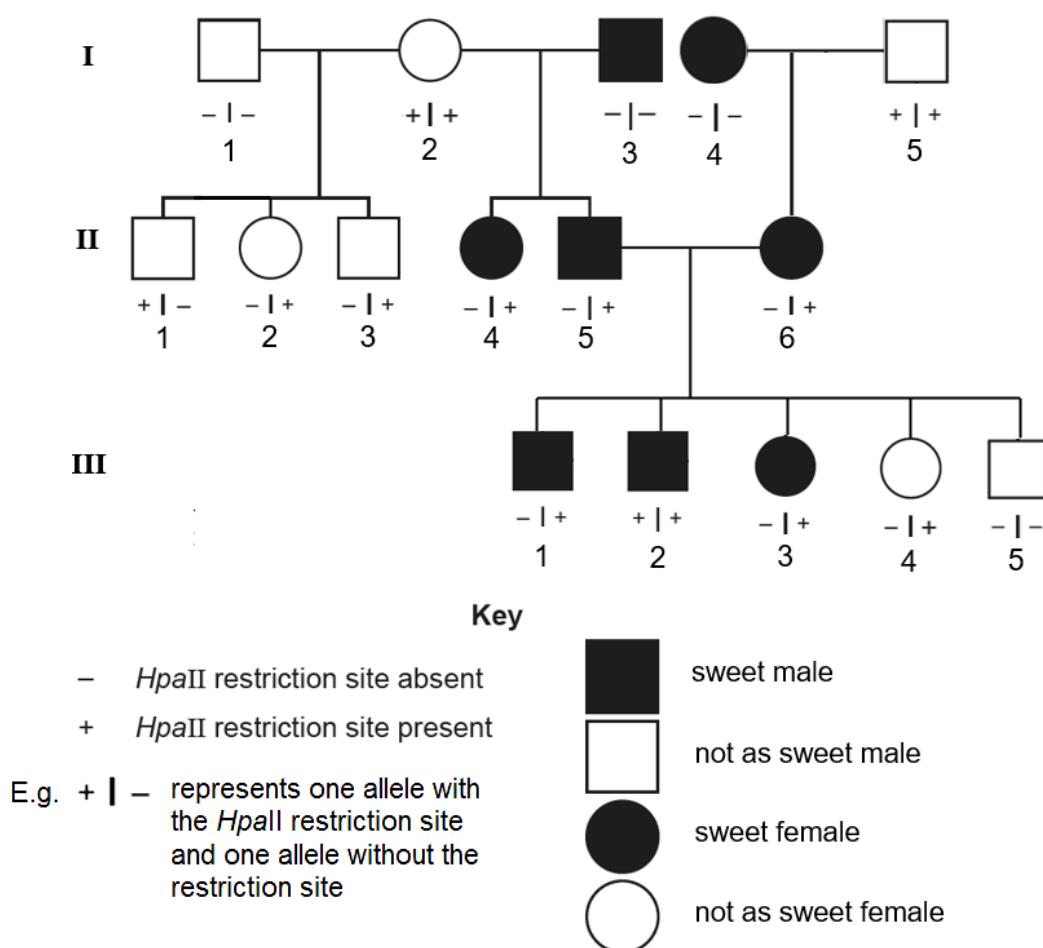


Fig. 7.1

(a) With reference to Fig. 7.1, identify an evidence that supports these statements:

(i) the sweet phenotype is due to a dominant allele.

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 [1]

(ii) the gene **R/r** is located on an autosome.

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 [1]

(b) The perceived positive correlation of flesh colour intensity and sweetness could be due to complete linkage between genes **R/r** and **T/t**. Using the symbols provided, draw a genetic diagram to show the expected phenotypic ratio for the cross between individuals II-5 and II-6 if genes **R/r** and **T/t** are completely linked.

[5]

[Turn over]

- (c) State the expected phenotypic ratio for the cross between individuals II-5 and II-6, if genes **R/r** and **T/t** segregate independently.

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..... [1]

- (d) The scientists aim to verify if there is a positive correlation between flesh colour intensity and sweetness.

A cross is carried out between two individuals which are heterozygous for both the genes **R/r** and **T/t**.

State the null hypothesis and statistical test that would need to be carried out in order to verify the correlation.

Null hypothesis:

.....

Statistical test: [2]

[Total: 10]

Question 8 starts on Page 22

[Turn over

- 8 The Galapagos marine iguanas (*Amblyrhynchus cristatus*) inhabit the coastlines of islands throughout the Galapagos archipelago. They mainly live in colonies on rocky shores, where they bask after foraging for underwater algae and seaweed in the relatively cold waters. The rocks on which they bask on are occasionally exposed to heavy waves. Like other reptiles, they are ectotherms with limited physiological means of keeping their body temperature constant and often rely on external sources of heat.

Morphological and genetic data have revealed that the closest relative of *Amblyrhynchus* is the genus of terrestrial iguanas, *Conolophus*, which is also endemic to the Galapagos.

Fig. 8.1 shows the overall morphology of the marine and terrestrial iguanas, as well as their claw structures.



Marine iguana



Terrestrial iguana



Marine iguana



Terrestrial iguana

Fig 8.1

- (a) Based on the information provided, explain how marine iguanas are better adapted to the marine environment compared to terrestrial iguanas.

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Scientists have observed 17 populations of the Galapagos marine iguanas, all belonging to the same species, on the different islands in the archipelago.

The *cytochrome b* (*cytb*) gene of the mitochondrial genome was analysed and compared across the 17 iguana populations. Based on the analysis, scientists were able to identify three major clades as shown in the phylogenetic diagram in Fig. 8.2.

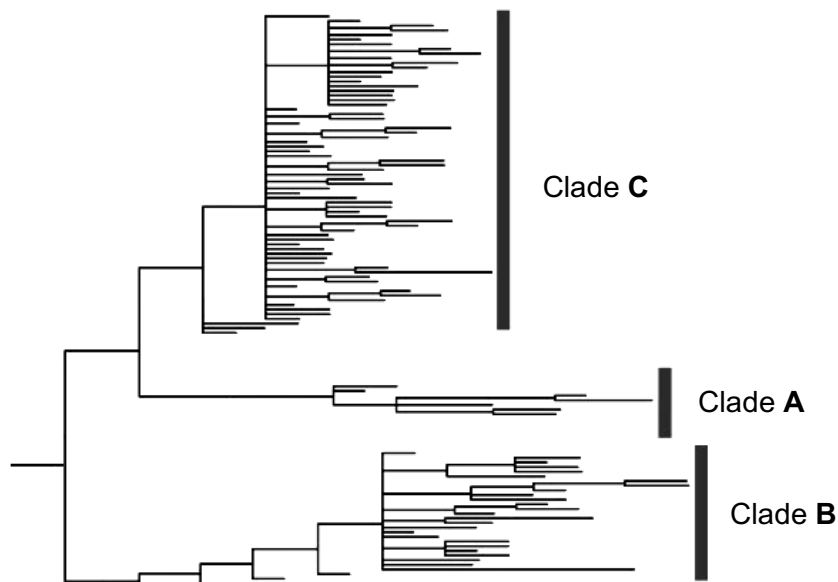


Fig. 8.2

- (b) (i) Define what is meant by a “clade” as shown in Fig. 8.2.

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 [1]

- (ii) Explain what the branch lengths of the phylogenetic diagram in Fig. 8.2 represent.

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 [2]

[Turn over

Fig. 8.3 maps the phylogeographic distribution of the three clades on the Galapagos islands, where the circles represent the proportion of the different clades of marine iguanas on each island.

It is known that the volcanic islands of Galapagos did not emerge at the same time, but progressively over a period of three million years. The map indicates the age of each island in mya (million years ago). The nearest mainland country, Ecuador, is 900 km east of these islands.

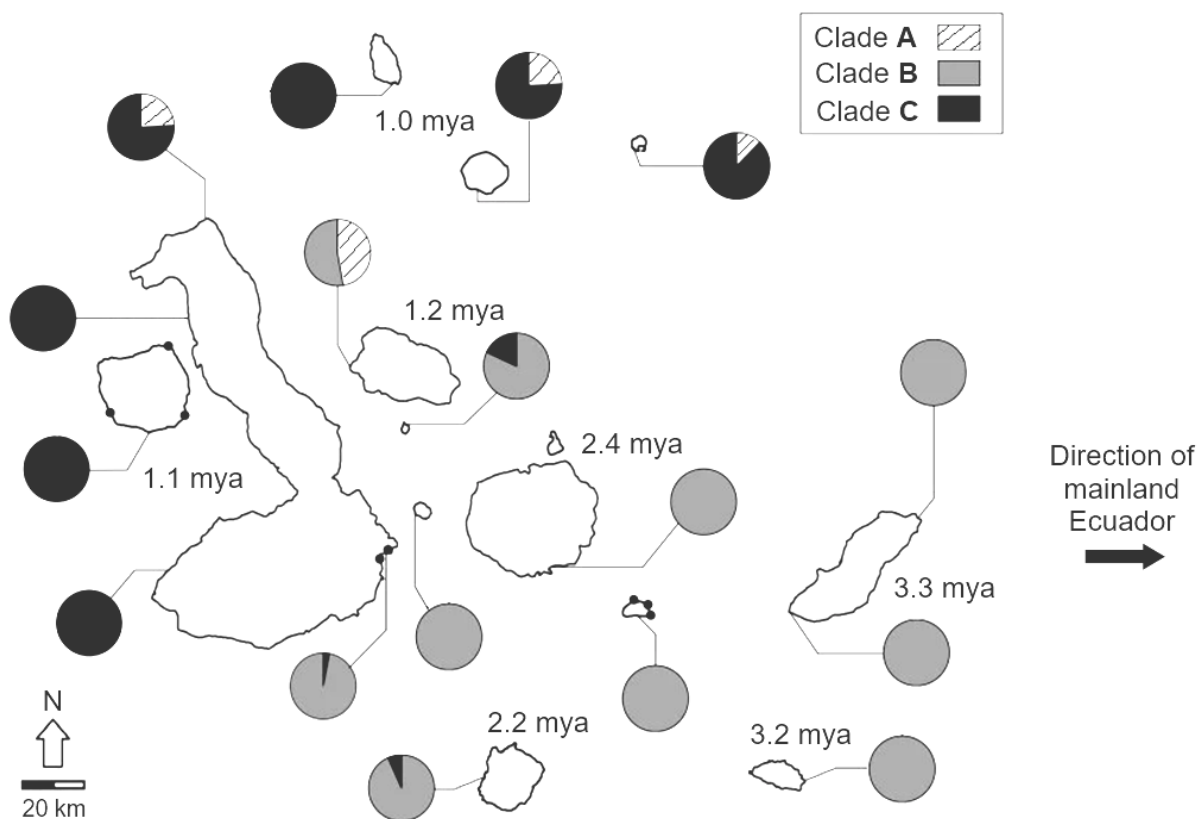


Fig. 8.3

- (c) (i) With reference to Fig. 8.3, describe how the phylogeographic distribution of marine iguanas (*A. cristatus*) belonging to clades **B** and **C** differ.

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- (ii) Suggest an explanation for the difference in phylogeographic distribution described in (c)(i).

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[Total: 10]

[Turn over

- 9 Glucose is an important substrate in cellular respiration. Fig. 9.1 shows the mTOR signalling pathway that drives cell growth by greatly stimulating glucose uptake and utilisation in a normal cell.

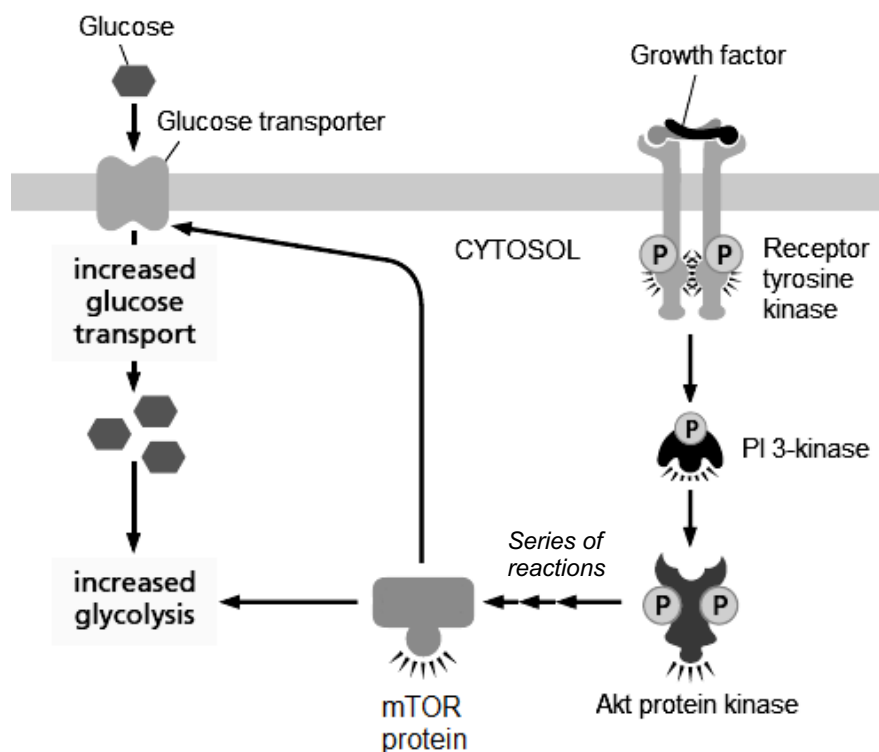


Fig. 9.1

- (a) (i) State one chemical property of the growth factor.

..... [1]

- (ii) Describe how the binding of PI 3-kinase to the receptor tyrosine kinase leads to the cellular responses in Fig. 9.1.

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The abnormal growth of cancer cells is highly dependent on increased glucose uptake and reliance on glycolysis for ATP production.

Recently, it has been suggested that having a ketogenic diet may be more inexpensive and easier to adopt than certain traditional anti-cancer therapies. A ketogenic diet is characterised by a high-fat and low-carbohydrate intake which is expected to decrease blood glucose levels and conversely increase ketone body levels. It is found that ketone bodies cannot be utilised by cancer cells.

Fig. 9.2 shows the effect of a ketogenic diet on a normal cell, where the decrease in glucose availability prevents some cellular processes from occurring.

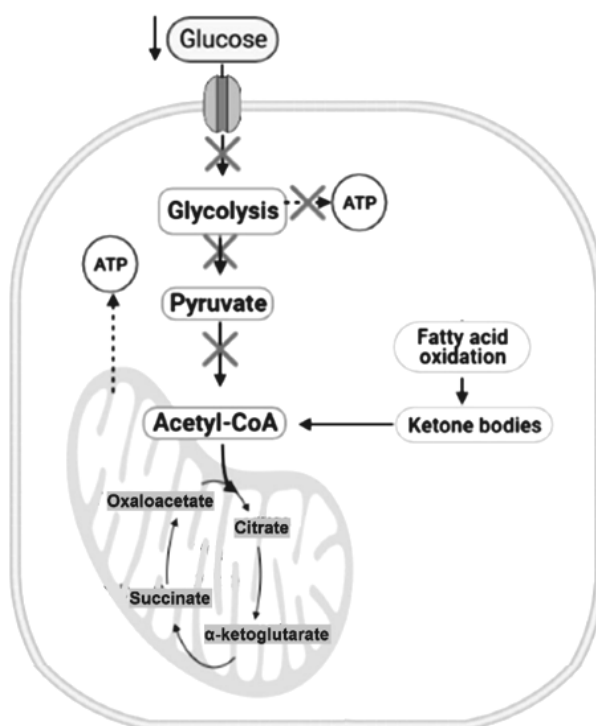


Fig. 9.2

(b) (i) Describe how a ketogenic diet could act as a possible anti-cancer treatment.

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[Turn over

- (ii) While ketogenic diets show promise in limiting cancer cell proliferation, there are also concerns regarding its usage over a longer period of time.

Suggest the possible implications of having a ketogenic diet in the long term.

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..... [2]

[Total: 10]

10 One distinctive feature of the adaptive immune system is the ability to generate millions of different antibody molecules to specifically target different foreign antigens.

- (a)** Fig. 10.1 shows the structure of a typical antibody, where two major segments of the antibody have been labelled with blank boxes.

Using the letters '**S**' to represent somatic recombination, '**H**' to represent hyper-mutation, and '**C**' to represent class switching, indicate in the blank boxes which segment of the antibody is affected by each of the three processes (each box may be filled by more than one symbol).

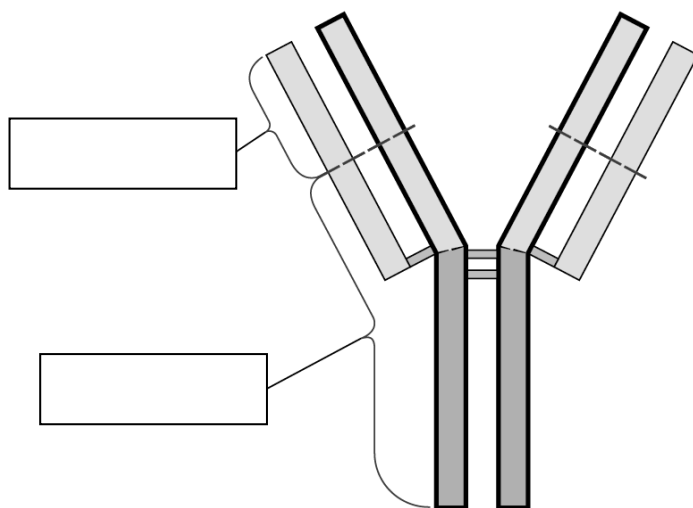


Fig. 10.1

[3]

[Turn over

Fig. 10.2 shows the concentrations of different antibody types following an initial exposure to a specific antigen and a repeated exposure to the same antigen.

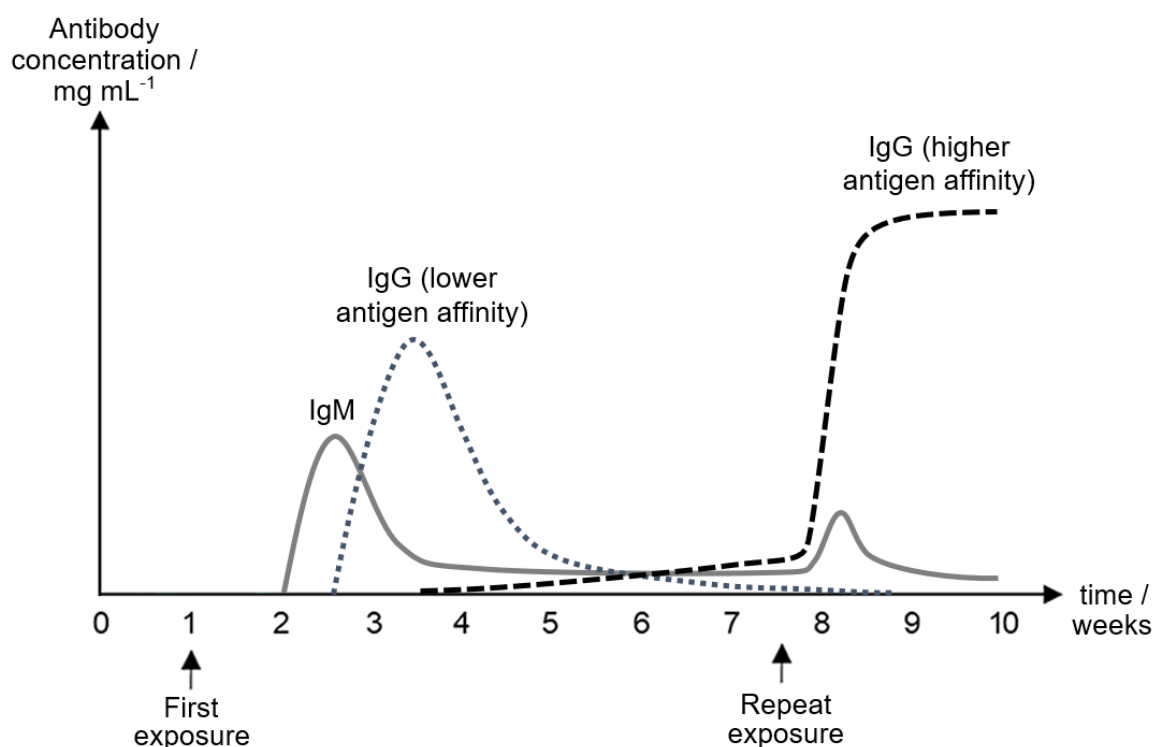


Fig. 10.2

- (b) With reference to Fig. 10.2, explain how the type of antibody secreted by B cells changes during the primary immune response.

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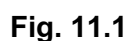
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[Total: 6]

- Fig. 11.1 shows the changes in global biodiversity and in the total area protected from 1965 to 2005, in terrestrial and marine habitats.



- (a) (i)** Compare the relationship between total area protected and global biodiversity in marine habitats:
- between 1970 and 1990, and
 - between 1990 and 2005.

..... [4

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- (ii) Suggest why a smaller area of marine habitats has been protected than of terrestrial habitats.

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..... [1]

[Total: 5]